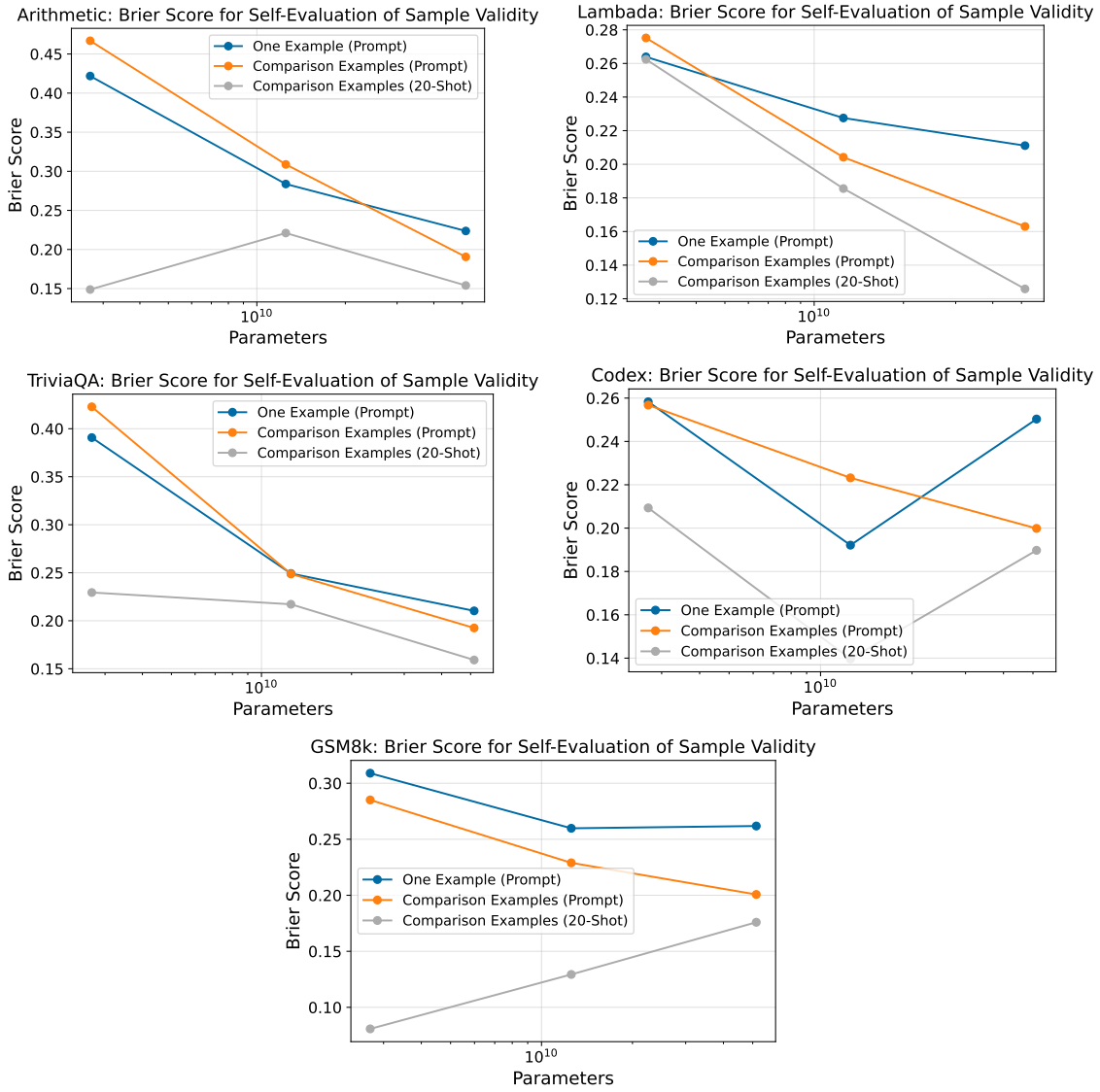

Figure 30 Here we show the Brier scores for model self-evaluation with three methods. Note that the Brier score combines accuracy of the True/False determination with calibration, and 20-shot evaluation with comparison samples performs best in every case. The model size trends in Brier score for Arithmetic, Codex HumanEval, and GSM8k may look surprisingly unpromising, but this is because smaller models have extremely poor performance on these tasks (see Figure 1), meaning that one can achieve a very low Brier score simply by predicting P(True) ≈ 0 uniformly.

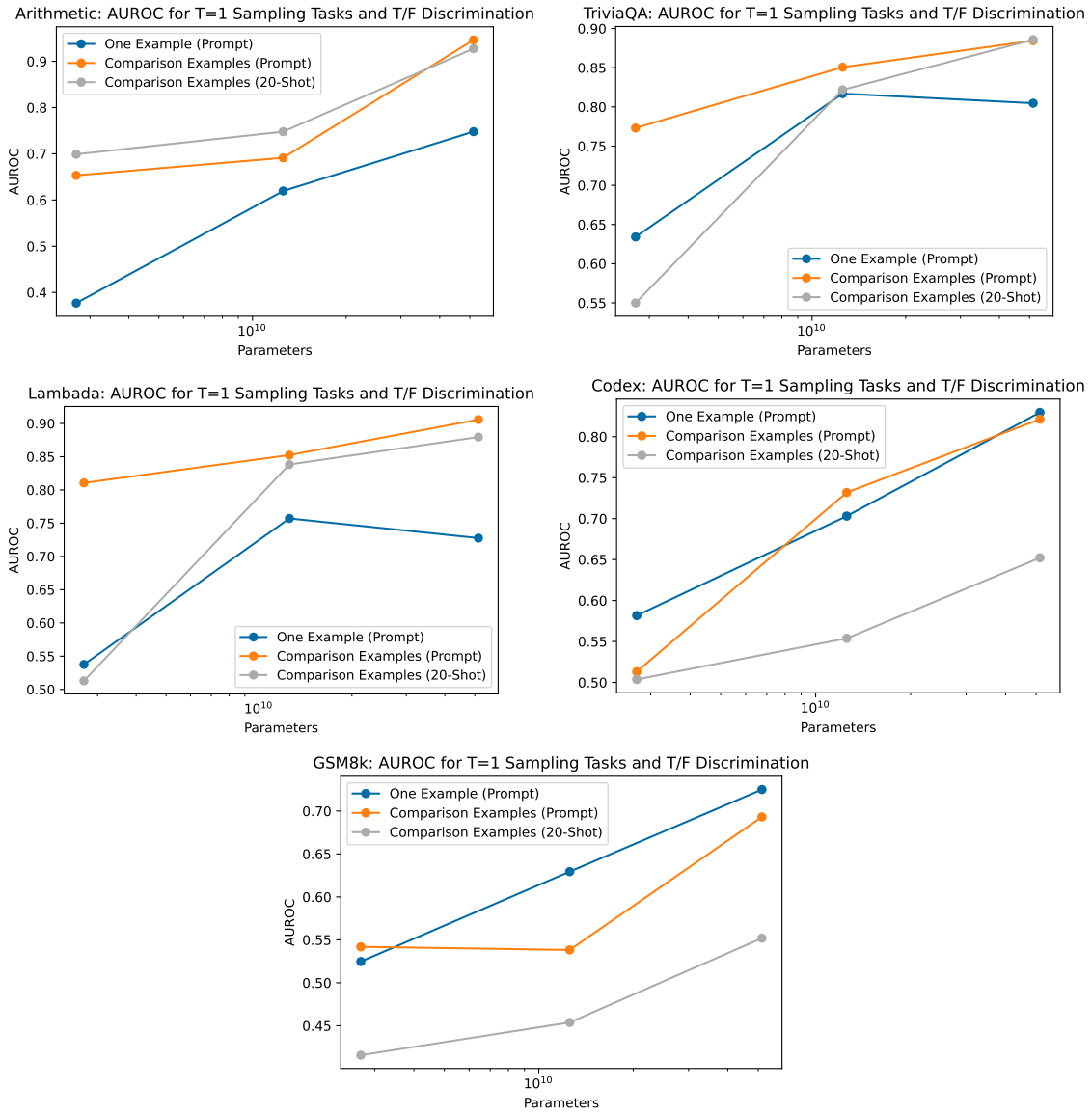

Figure 31 We show the AUROC scores for P(True) with several different methods – a fixed prompt and basic self-evaluation, a fixed prompt and self-evaluation with comparison examples, and 20-shot with comparison examples.

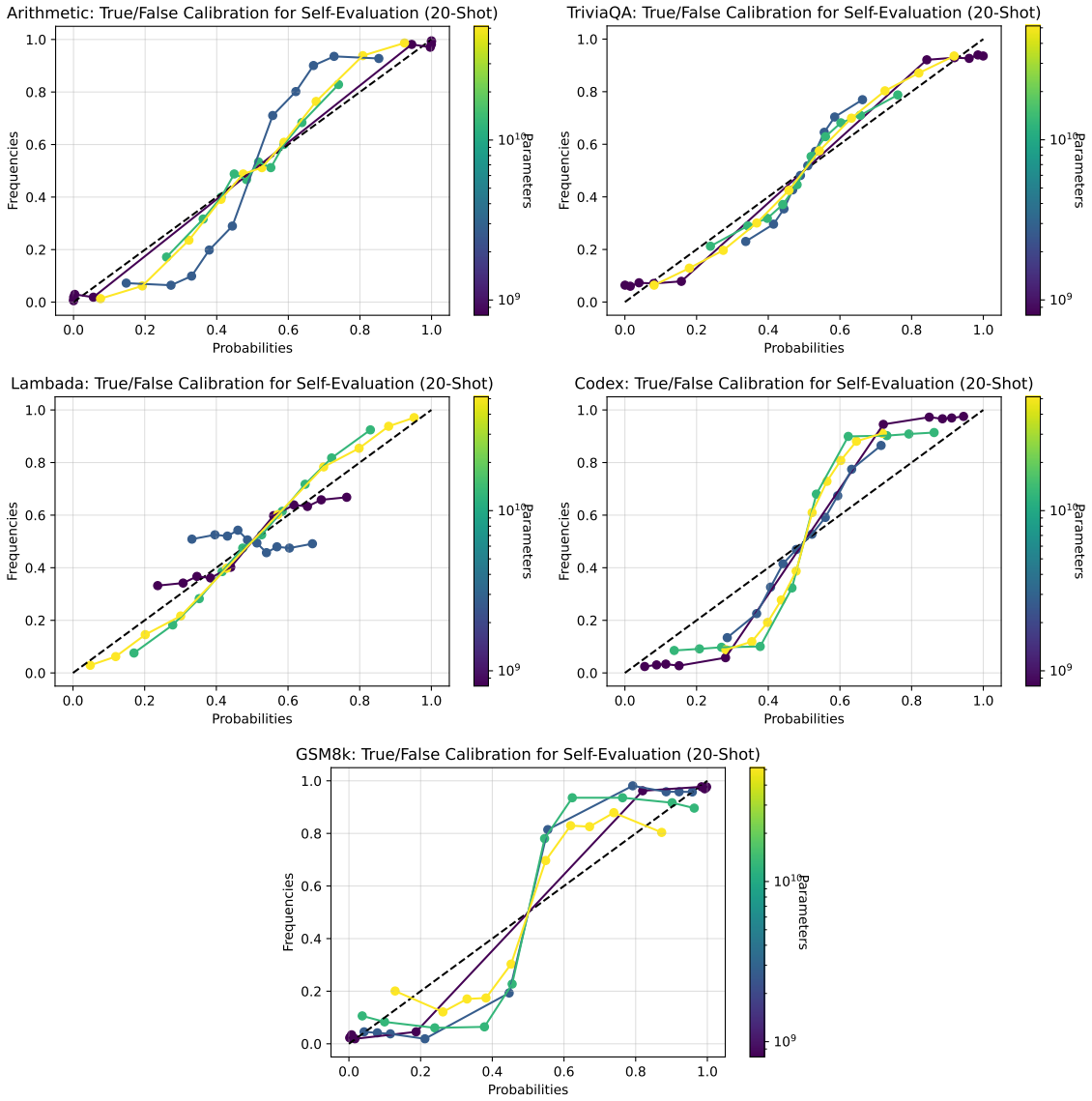

Figure 32 We show the calibration curves when models self-evaluate the likelihood that their own samples are in fact correct. It is worth noting that for some of the evaluations (GSM8k, Codex, Arithmetic), smaller models get almost every question wrong, which means that in these cases the small models are only well-calibrated because they are making trivial predictions.